

Impact of nanoparticle movement on oil mobilization: Particles adsorbed at the oil–water interface versus particles dispersed in bulk fluid

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ABSTRACT

Active colloid particles have been used in many fields due to their good performance in enhancing mass transport or modifying interfacial energy. Previous studies have shown that both nanoparticles adsorbed at oil–water interface and those dispersed in bulk fluid could enhance oil recovery during the oil production process. It is challenge to directly compare these results and clarify the effect of nanoparticle's location on oil detachment from rock surface because different kinds of nanofluids were used in those work. Herein, the oil mobilization performance by nanoparticles adsorbed at oil–water interface and dispersed in the bulk phase were investigated from the pore-scale to core-scale by using one material. Two fluorescent quantum nanodots with different affinities for oil and water phases were synthesized using the same materials but with different compositions and solvents. Their locations were observed under the laser confocal microscopy. The nanoparticles adsorbed at the oil–water interface not only presented a lower interface tension, but also demonstrated a notable capability to change the wettability from oil-wet to water-wet due to its strong tendency to rock surface. The particles-laden interface can also increase the stability and shear viscosity of formed emulsions. Furthermore, it featured a larger swept area of 6.6 % in microfluidic flooding and a higher displacement efficiency of 1.7 % in the core flooding experiments. All these findings indicate that the particle-adsorbed at the oil–water interface is more favorable for enhanced oil recovery.

1. Introduction

Nanoparticle movement in liquid–liquid system has garnered significant research interests due to its practical applications in various fields, including drug encapsulation, industrial emulsification, cosmetics, and oil production [1–5]. Among these applications, its potential significance in the oil production from underground reservoirs is particularly noteworthy due to the substantial demand. Globally, there are more than 60 % of the remaining oil after the primary and secondary recovery in the worldwide, emphasizing the critical role of enhanced oil recovery (EOR) technologies [6,7]. Given the unique surface properties of nanomaterials, their addition to the aqueous phase has been suggested as a promising technique for EOR [8–10].

Both organic and inorganic nanomaterials have been tested for oil production improvement at the laboratory. The performance of nanoparticles and nanosheet in EOR has been scrutinized through various studies utilizing visualized micromodel and core flooding experiments.

The observed incremental recovery exhibits a wide range. In a comprehensive analysis, Ding et al. [11] collected 240 experimental data points from 39 publications and reported that the majority of incremental recovery achieved by nanomaterials was lower than 10 %. On the other hand, the study conducted by Roustaei et al. [12] demonstrated that their used nanomaterials could reduce oil–water interfacial tension (IFT) to 1.95 mN/m, resulting in a substantial incremental oil recovery of 25.4 %. Another investigation by Luo et al. [13] revealed that increments as high as 15.2 % could be achieved with the injection of 3–4 pore volume (PV) of nanomaterials after water flooding. Interestingly, even within a single study, the same nanomaterial contributed to different incremental oil recovery by considering various concentrations and core permeabilities [14]. Understanding the intrinsic reasons for these variations has become a subject of contemplation.

Structural disjoining pressure, reduction in interfacial tension (IFT) between oil and water, wettability alteration of rock surface and stabilization of emulsion formation are four widely reported EOR

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mechanisms of nanoparticles. The affinity of nanoparticles to partition at liquid–liquid interface and solid–liquid surface was considered as a pivotal factor that significantly affects the efficacy of these mechanisms. Unlike the amphiphilic structure of surfactants, there is no micelle formation for the nanoparticles [15]. The position of nanoparticles at oil–water interface is governed by their sizes and surface wettability [16,17]. The mechanism of wettability alteration by nanoparticles is different from that by surfactant. Based on the Young's law, surfactant could change the surface tension of oil–rock or water–rock to make wettability changing [18]. The rock is oil wet when the surface tension of oil–rock is lower than that of water–rock and vice versa. The IFT between oil and water could not change the wettability but affect the wetting degree. The nanoparticles can adsorb on the rock surface and changing the wettability of rock surface based on the nanoparticles' properties [19,20]. Its wettability alteration ability is related to the salinity, ion type and nanoparticle concentration [21,22]. The nanoparticles of intermediate wettability usually featured with a large free energy of adsorption, which presented a stronger tendency for the aggregation at the oil–water interface [23]. But there is no clear evidence to confirm whether the nanoparticle's position has close relationship with the oil mobilization in porous medium.

Studies by Wasan and Nikolov [24] reported that nanoparticles adsorb to the solid surface, leading to the spreading of immiscible liquids. Other researchers have also observed that the adsorbed nanoparticles can alter the wettability [25]. In addition to the nanoparticle adsorb on the solid–liquid surface, the presence of particle-adsorbed at the oil–water interface is considered a critical property for EOR. Yin Wu et al. [26] proposed that amphiphilic nanoparticles spontaneously accumulating at the oil–water interface could result in an incremental oil recovery of more than 25 % after water flooding. Wu et al. [27] and Liu Wu et al. [28] both found the nanoparticles adsorbed at the oil–water interface could help reducing the interfacial tension between oil and water, facilitating the mobilization of trapped oil. Moreover, the particles adsorbed at the oil–water interface formed Pickering emulsions, enhancing interfacial film strength due to the high mechanical stability [29–31]. Pickering emulsions hold significant potential for application in enhanced oil recovery for high salinity and heterogeneous reservoirs [32–34].

All the aforementioned studies suggest that particles adsorbed at the oil–water interface contribute significantly to oil mobilization. However, there is also evidence indicating that some fully water-wet nanoparticles can exhibit high efficiency in the oil displacement experiments [35–37]. Aqueous suspension of nanoparticles has been reported to be effective in both reducing IFT and stabilizing emulsions. Studies by Biswal et al. found the nanoparticles influence the surface excess concentration of surfactants. The increased concentration of surfactants at oil–water interface, rather than nanoparticles themselves, leads to reduce IFT [38]. Pilapil et al. proposed that non-interfacially adsorbed nanoparticles could stabilize emulsions by enhancing forces between dispersed oil droplets [39]. Notably, there is a lack of comparative studies on how does particle movement specifically affect oil recovery from the porous media within the literature. The diversity of nanofluids used in these studies make direct comparisons challenge.

The aim of this work is to investigate the impact of the location of particles on oil mobilization via the utilizing synthesized carbon quantum nanodots with different surface characteristics under the help of surfactant. The locations of two quantum nanodots on oil phase were observed using a laser scanning confocal microscope. Their interactions with kerosene and rock surface were analyzed through assessments of emulsion stability, rheology and changes in rock wettability. The study further delves into their influence on the oil mobilization by analysis of residual oil displacement from the pore scale to core scale. The findings of this work aim to establish a connection between microscopic nanoparticles distribution and macroscopic oil mobilization. It will not only provide the insight on how to select nanofluids for oil production, but also help people understand the effect of solid movement on oil

detachment at the confined space.

2. Materials and methods

2.1. Materials

The citric acid ($C_6H_8O_7$, 99.5 %), tobas acid ($C_{10}H_9NO_3S$, 98 %), and ethylene diamine ($C_2H_8N_2$, 99 %) used for quantum dots synthesis were purchased from Shandong Xiya Chemical Industry Co. LTD. The other raw materials, m-phenylenediamine ($C_6H_8N_2$, AR) and sodium chloride (NaCl) were purchased from Tianjin Fuchen Chemical Reagent Co. LTD. The ethanol (C_2H_5OH , AR) acting as solvent was purchased from Tianjian Jingdongtianzheng Precision Chemical Reagent Factory. Deionized water produced by BK-10B from Qianjing Environmental Equipment Co. Ltd was used in all the experiments. Sodium dodecyl benzene sulfonate (SDBS, Sigma) was used as a water-soluble nonionic surfactant. It has a molecular weight of 348.5 g/mol and a hydrophilic-lipophilic balance (HLB) of 10.6. Kerosene purchased from Tianjin Damao Chemical Reagent Co. LTD., acted as the oleic phase in both interfacial properties tests and emulsion preparation. Crude oil from Karamay oilfield with a viscosity of 7.5 mPa·s at room temperature (25 °C) was used to soak core slices for wettability change test. To distinguish the oil phase and aqueous phase in oil displacement experiments, kerosene and crude oil with a volume ratio of 1:1 were mixed to saturate micromodels and core samples. All chemicals are of analytical grade for direct use without further purification.

2.2. Synthesis of quantum dots (CDs)

High-yield quantum dots (CDs) were synthesized by using the hydrothermal method. 60 μ L of ethylene diamine and 0.735 g of citric acid as precursors, and deionized water (25 ml) serving as the solvent. The homogenized precursor solution was obtained by ultrasonic for 20 min and reacted in the Teflon reactor at 160 °C for 12 h. After this preparation process, the B-CDs were prepared as shown in Fig. 1(a). 0.06 g tobas acid and 0.2g m-phenylenediamine were added into 20 ml ethanol solution to obtain the mixed solution. Then mixed solution was transferred to a Teflon reactor. After 10 h of reaction with the 200 °C environment temperature, the G-CDs were prepared, as shown in Fig. 1(b).

2.3. Characterization of CDs

The morphology and microstructure of the CDs were examined using Transmission electron microscopy (TEM) and High-Resolution Transmission Electron Microscopy (HRTEM) on JEOL JEM-2100 with an accelerating voltage of 200 kV. The atom force microscope (AFM) (Bruker Scan-Dimension-Icon System) was used to study the surface of the microstructure. The Fourier Transform Infrared (FTIR) spectrum was measured on Bruker Vertex 70v to characterize the functional group from 500–4000 cm^{-1} on the surface of CDs. UV–Vis absorption measurements were conducted using a fluorescence spectrometer (FS-05) from Techcomp (China Ltd).

2.4. Interfacial properties of CDs

The contact angle of rock-CDs solution- kerosene was measured using the pendant drop method on a force tensiometer (Krüss K100, German) by dropping the kerosene on a rock slice substrate which immersed in 3 wt% concentrations of B-CDs and G-CDs solutions. The contact angle of rock- CDs- bine was also measured for comparison. The polished rock slices were aged in the crude oil for one week to make them oil-wet before use. The interfacial tension between two combination solutions, containing 3 wt% concentrations of CDs and 1 wt% concentrations of SDBS, and kerosene, was measured using a Texas-500C spinning drop interface tensiometer (CNG company, US) with a

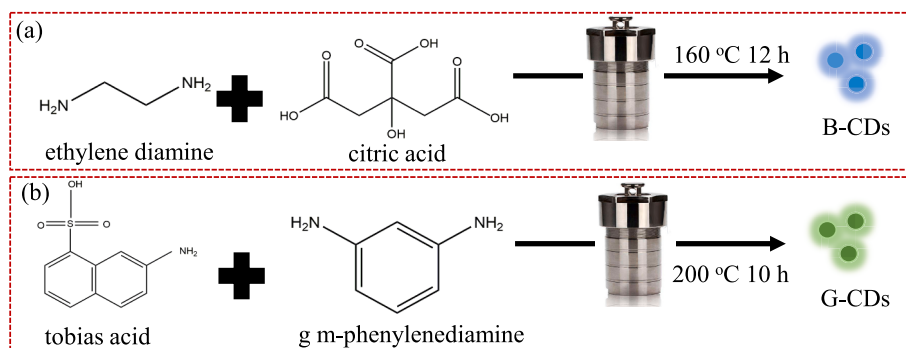


Fig. 1. Schematic diagram of the quantum dots synthesis.

rotation speed of 3000 rpm. All the aqueous phase contained 1 wt% of sodium chloride (NaCl) to promote the interfacial activity of surfactant.

2.5. Emulsion preparation and its rheology

Three types of solution, namely, 1 wt% SDBS, a combination of 1 wt% SDBS and 3 wt% B-CDs, and a combination of 1 wt% SDBS and 3 wt% G-CDs were used as aqueous phase under a NaCl concentration of 1 wt%. Kerosene was added to the aqueous phase to achieve the desired water–oil volume ratio of 1, and the mixed solution was stirred using a homogenizer at 100 rpm for 10 min. The shear viscosities of three emulsions were recorded by a HAAKE RheoStress 6000 rheometer (Thermo Fisher Scientific, US).

2.6. Confocal laser scanning microscopy (CLSM)

An Olympus IX83 inverted microscope equipped with the Yokogawa CSU-X1 spinning disk confocal unit and a 405 nm continuous wave laser (Coherent Inc.) was used to capture the distribution of CDs in the emulsions. Since only CDs emit under the laser, their positions can be easily observed by CLSM.

2.7. Oil mobilization in a microfluidic device

Oil displacement in the micromodels were conducted for the comparison of oil mobilization difference between two CDs combined SDBS solutions. The experimental setup of the microfluidic platform was shown in Fig. 2a. It included a color camera (Sony-A7S2) attached to a stereo zoom microscope (Zeiss-Discovery StereoV12), a syringe pumps (Harvard PHD Ultra 2000), and a microchip. The camera can real-time record the images, which were used for oil mobilization analysis via the later image recognition. Polydimethylsiloxane (PDMS) was used to fabricate the oil-wet microfluidic chip using standard soft-lithography techniques. The pore structure of the microchip was designed based

on the real cast thin sections, with a pore size distribution ranging from 3 μm to 45 μm , as shown in Fig. 2b. Detailed parameters of this microchip can be found in our previous work [40]. The microchip was saturated with compound oil, a mixture of crude oil and kerosene at a volume ratio of 1:1. Each CDs with the concentration of 3 wt% combined with 1 wt% of SDBS, was separately injected into the microchips for oil displacement at a constant flow rate of 1 $\mu\text{L}/\text{min}$. The displacement continued until no color change was observed. Oil recovery was identified through pixels changing using Image-J software. The oil mobilization mechanisms of the two CDs with different interface properties were compared through the analysis of the pore-scale remaining oil distribution.

2.8. Core flooding experiments

Three outcrop cores with a diameter of 2.5 cm and a length of 30 cm were used for core-scale oil displacement experiments. They featured with the permeability of 350 mD and a porosity of 26.5 %. The core was placed in the core holder. The inlet was connected with the injection pump and the outlet was connected with the measuring system. The pressure sensor at the inlet will record the injection pressure during the displacement process. The confining pressure of the core holder was set 2 MPa higher than the injection pressure. The experimental procedures were as follow: vacuumize the core and saturate 1 wt% of sodium chloride (NaCl) brine, the compound oil, similar to the microfluidic experiments, was injected into the outcrop cores at a flow rate range from 0.1 to 1 mL/min to create initial oil saturation. Two CDs solutions with the concentration of 3 wt% combined with 1 wt% of SDBS and one solution with only 1 wt% of SDBS were separately injected into the outcrop cores for oil displacement until no oil coming out. The pressure and effluent were recorded to compare the performance of CDs in oil mobilization at core-scale.

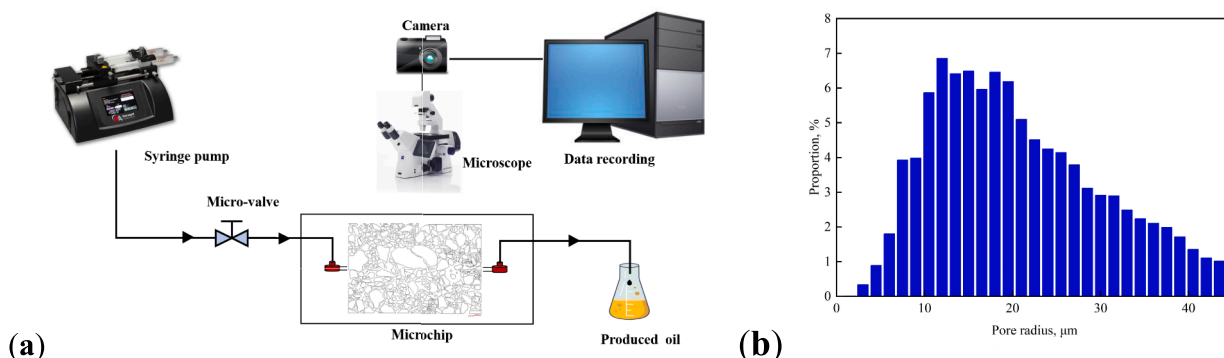


Fig. 2. (a) Microfluidics experimental setup and (b) Pore-throat distribution of microchip.

3. Results and discussion

3.1. Micromorphology of CDs

The geometrical characteristics of CDs were observed by TEM and AFM. The uniform dispersion of B-CDs and G-CDs in Fig. 3a and 3d indicated that they can dissolve well in the corresponding solvent. Their size distributions, calculated from the TEM micrograph via the image analysis in Fig. 3b and 3e, showed that the size range of both CDs was from 2.0 nm to 5.0 nm. The mean size of B-CDs, at 3.0 nm, was slightly smaller than that of G-CDs, at 3.75 nm. The thicknesses of B-CDs and G-CDs, investigated by AFM images in Fig. 3c and 3f showed that the average thicknesses were both less than 4 nm, further demonstrating the synthesized CDs were quasi-spherical. These regular-shaped nanodots could provide qualified materials for the following research.

3.2. Physicochemical characteristics of CDs

FTIR and UV-Vis absorption analysis were carried out to investigate the surface chemical bonds and internal electron conjugation of the CDs. In Fig. 4a, the FTIR spectrum of B-CDs showed several peaks at 3000 cm^{-1} , 1650 cm^{-1} and 1500 cm^{-1} , respectively attributed to C—H and C=O stretching vibrations [41,42], aromatic compound [42]. The peaks at 3500 cm^{-1} and 3300 cm^{-1} in FTIR spectrum of G-CDs in Fig. 4d were attributed to N—H and O—H, contributing to increased solubility [43]. Peaks at 1730 cm^{-1} and 1380 cm^{-1} were caused by C=O and C—O stretching vibrations [44,45]. The UV-Vis absorption spectrum of B-CDs in Fig. 4b showed an absorption peak at 260 nm attributed to $\pi\text{-}\pi^*$ transition, indicating the presence of aromatic compounds, consistent with the FTIR results. The strong absorption peak at 280 nm in Fig. 4e resulted from the $n\text{-}\pi^*$ transition, indicating unsaturated groups in G-CDs [46].

Fig. 4c and 4f presented the photoluminescence spectra, showing the fluorescence characteristic of B-CDs and G-CDs. The optical excitation wavelength and emission wavelength for B-CDs were at 390 nm and 440

nm, while for G-CDs, they were at 460 nm and 515 nm. The emission peaks of both CDs did not shift with the changing excitation wavelength, indicating excitation-independent properties. Additionally, the lifetime spectrum and stability results under various PH values and temperatures in Fig. S1 and Fig. S2 verified the good properties of both CDs. These results indicate that B-CDs and G-CDs can be utilized in the subsequent studies on oil–water interface behavior.

3.3. Effect of CDs on the contact angel and interface tension

The liquid–liquid–solid interface properties can be reflected by the contact angles test. The contact angle images and their comparison in Fig. 5a–d indicate that the added CDs make the wettability of rock surface change from oil wet to water wet, which is similar to the literatures' reports on the wettability changing by nanoparticles.

Compared to B-CDs, G-CDs could let the contact angle decrease by more than 10 degrees, presenting a better wettability change property. It indicates the amphiphilic nanoparticle has a strong tendency to move toward rock surface. The adsorbed nanoparticles make the wettability of rock surface change. The IFT curves of three systems with kerosene in Fig. 5e also show that G-CDs are featured with a low IFT value. Although the main components of two CDs are similar, the selection of synthetic raw materials and solvent cause different interface properties between G-CDs and B-CDs. Especially, the organic solvent of ethanol was used for G-CDs synthesis, making it present better wettability change and IFT reduction performance than B-CDs.

3.4. Effect of nanoparticle location on the bulk rheology

The fluorescent quantum dots will appear colorful under the confocal microscope. Since the wavelength of confocal microscope corresponds to blue, the blue areas in Fig. 6a and 6b reflect the location of CDs. Fig. 6a vividly shows that most B-CDs are dispersed in the aqueous phase, whereas the blue film in Fig. 6b indicates that G-CDs can adsorb at the oil–water interface. The different locations of two CDs provide the

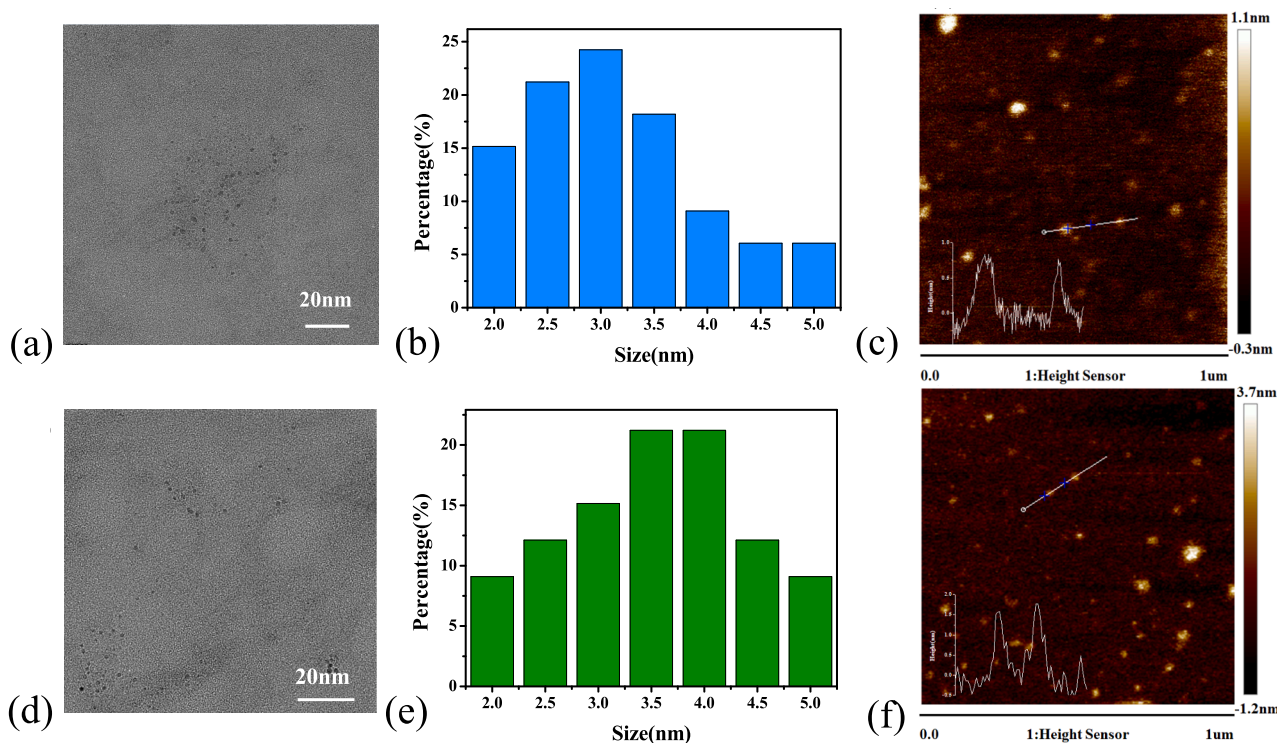


Fig. 3. TEM micrograph of (a) B-CDs and (d) G-CDs. The histogram of the size distribution of (b) B-CDs and (e) G-CDs. The AFM image of (c) B-CDs and (f) G-CDs (the insert: the thickness on the white line).

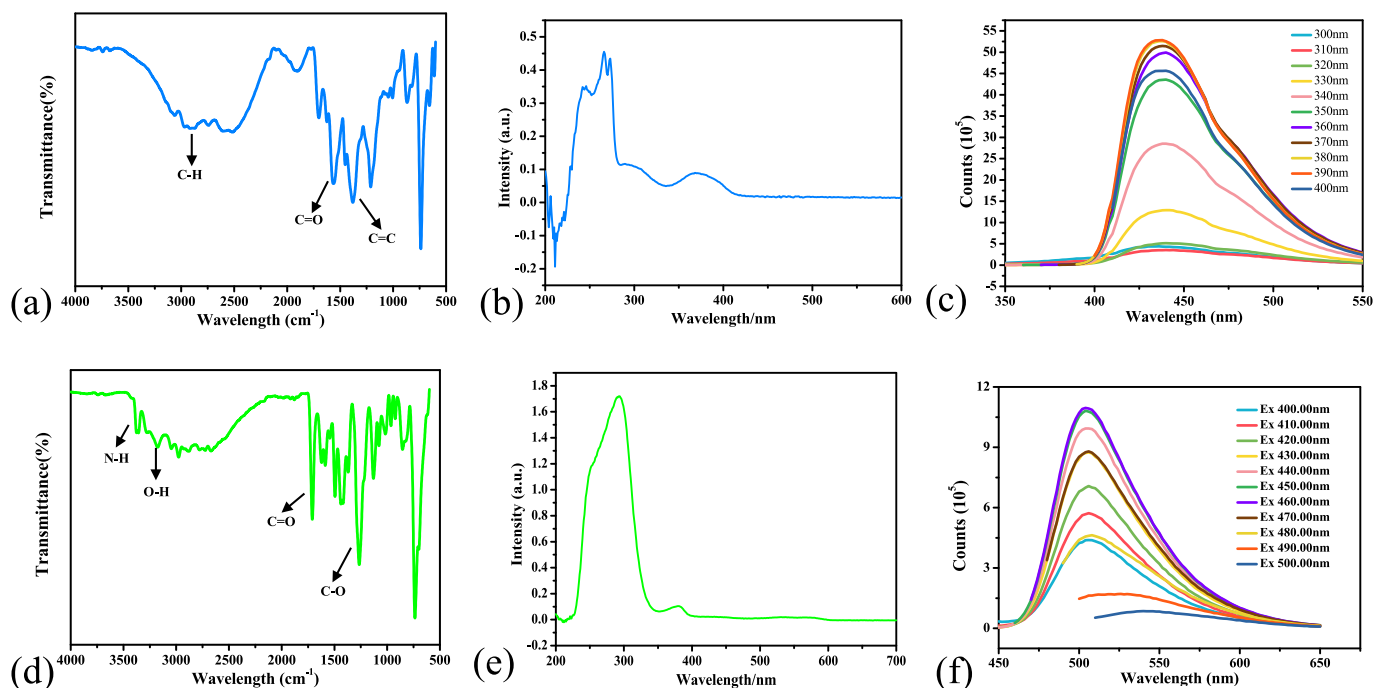


Fig. 4. Physicochemical characteristics of CDs. The FTIR spectrum of (a) B-CDs and (d) G-CDs. UV-Vis spectrum of (b) B-CDs and (e) G-CDs. The photoluminescence spectra of (c) B-CDs and (f) G-CDs.

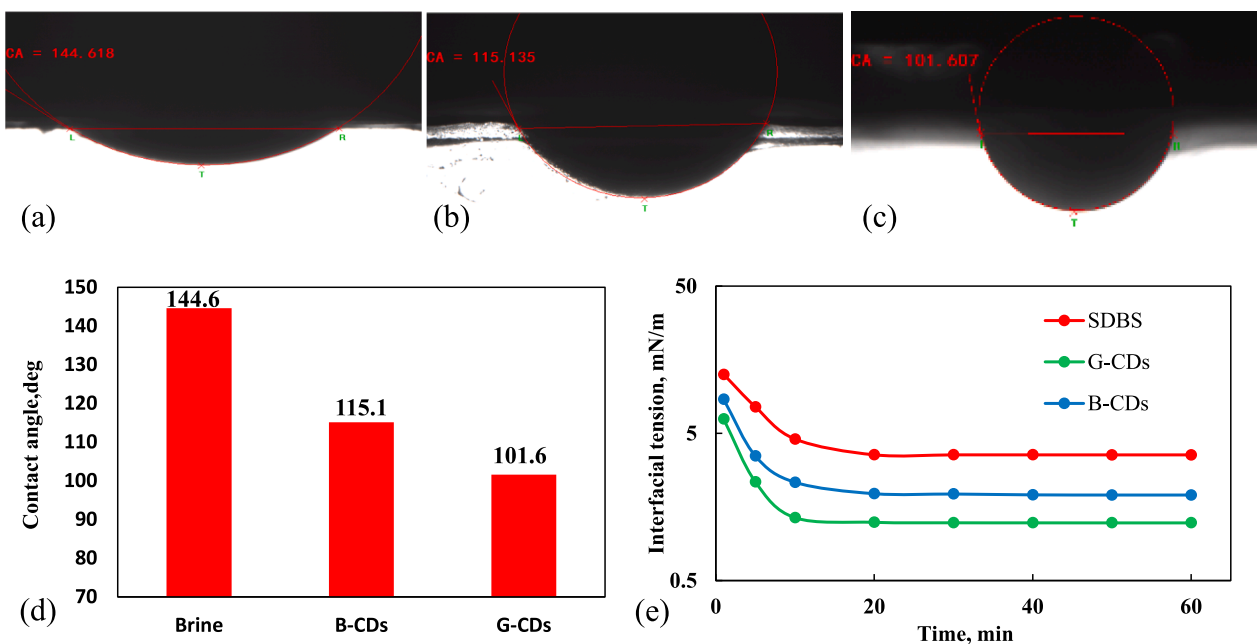


Fig. 5. Effect of CDs on kerosene-brine-rock and liquid-liquid properties. The contact angle images of (a) brine, (b) B-CDs and (c) G-CDs. (d) Comparison of contact angle of three systems. (e) IFT between kerosene and the solution of SDBS, B-CDs and G-CDs.

possibility for investigating the effect of nanoparticles movements on the oil mobilization in the subsequent work.

The shear viscosity of prepared emulsions using surfactant adding CDs and surfactant alone in Fig. 6c is influenced by nanoparticle location, as it can increase the viscosities of emulsions wherever the nanoparticles are present. Unlike emulsions stabilized by surfactants via the IFT reduction between the phases, the enhanced interface film rigidity by CDs could significantly improve the stability of emulsions [29–31,47]. This is the reason for the viscosity of emulsions prepared with CDs being higher than that of emulsions prepared without CDs at

the various shear rates. The difference of shear viscosity is more noticeable at both low and high shear rates. Comparing the shear viscosity between the emulsions prepared with B-CDs and G-CDs, the adsorbed nanoparticles on the interface could increase the interfacial viscosity and inhibit droplets coalescence [48–51], making G-CDs emulsions present a higher viscosity across the entire tested shear rate range, especially for the shear rates higher than 100 s^{-1} .

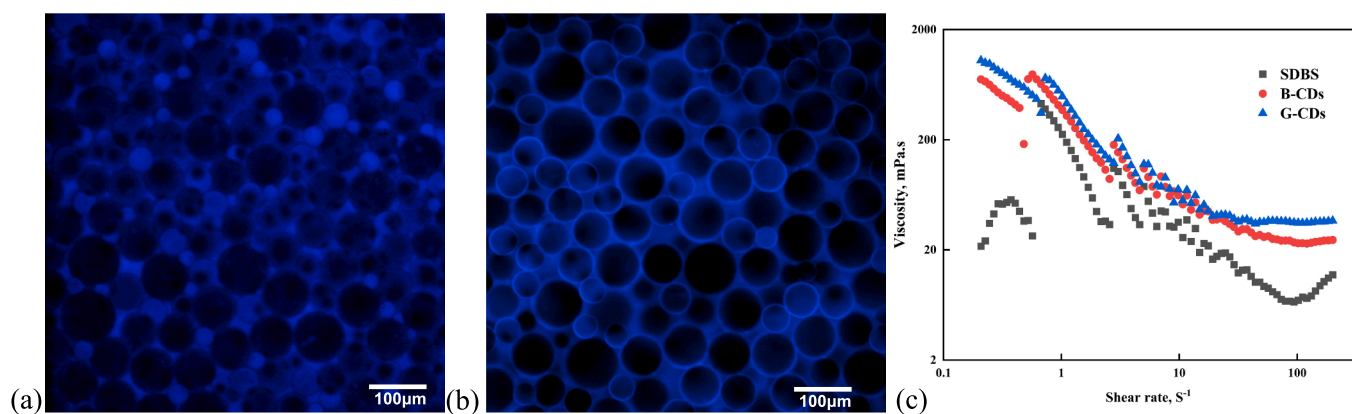


Fig. 6. Confocal images of emulsions that prepared using (a) B-CDs and (b) G-CDs. (c) Rheology properties of emulsions which prepared with SDBS alone and SDBS adding two CDs.

3.5. Pore-scale remaining oil mobilization using two CDs

The comparison of two CDs on the oil mobilization can reveal their effect on the interaction mechanism of oil–water–solid surface in the flowing state. Fig. 7a and 7c display the oil displacement results of two CDs with SDBS in the PDMS microchip. Since the micromodel is square, the injected fluid could not sweep the whole areas. The recovery factor of the micromodel was simultaneously determined by the swept area and displacement efficiency from the pore scale. Therefore, the EOR mechanism of nanofluids could be summarized from the swept area enlargement and oil displacement efficiency improvement in this experiment. The dark color in the microchip is residual oil. The residual oil after G-CDs with SDBS flooding seems less than that after B-CDs with SDBS flooding. To highlight the distribution and mechanism formation of residual oils, they were classified into four types based on their morphologies and positions. The detailed classification method was reported in our previous work [52,53]. The pore throat residual oil refers the snapped oil droplet that was stuck in the narrow throat. The hydrodynamic residual oil formed as the injected fluid could not contact the oil, usually existing in the dead end and corner of the pore. The micro heterogeneous residual oil is the continuously distributed oil that failed to be swept by the displacing fluid. This type of residual oil is formed due to the non-uniformity of the pores and is widely present throughout the displacement process. The last type of residual oil is the oil film, indicating oil attached the solid surface controlled by the wettability. The hydrodynamic residual oil and micro heterogeneous

residual oil are dominated by the swept volume enlargement ability of the displacing fluid. The formation of pore throat residual oil is related to interface tension between the displacing and displaced fluid. The oil film can be mobilized by changing the wettability of rock surface from oil-wet to water-wet. For example, the amount of oil film after one fluid displacement, the Therefore, the amount of these residual oils could indicate the oil mobilization mechanisms of displacing fluids.

The changes in the amounts of residual oil during the CDs injection in Fig. 8 can help understand the oil displacement mechanisms differences between B-CDs and G-CDs. The micro heterogeneous residual oil decreases, while the pore throat residual oil and oil film increase during two CDs injection. It indicates that micro heterogeneous residual oil can be converted into other residual oils as the displacement proceeds. However, it still dominates the whole displacement process. The hydrodynamic residual oil slightly increases, but its amount is obviously less than others. Compare the oil mobilization differences between G-CDs injection and B-CDs injection in Fig. 8a and b, the relative amount of micro heterogeneous residual oil for G-CDs is less than that for B-CDs, which means the G-CDs featured with a considerable ability for swept areas enlargement. The larger relative amount difference between pore throat residual oil and oil films for G-CDs indicates more oil could be mobilized via the mechanism of wettability changing rather than IFT reduction. The co-adsorption of G-CDs and SDBS over the rock surface make their nanoaggregates featured with a better wettability alternative performance [22]. This result is consistent with the wettability changing ability of G-CDs and B-CDs in Fig. 5. It indicates the added SDBS may not

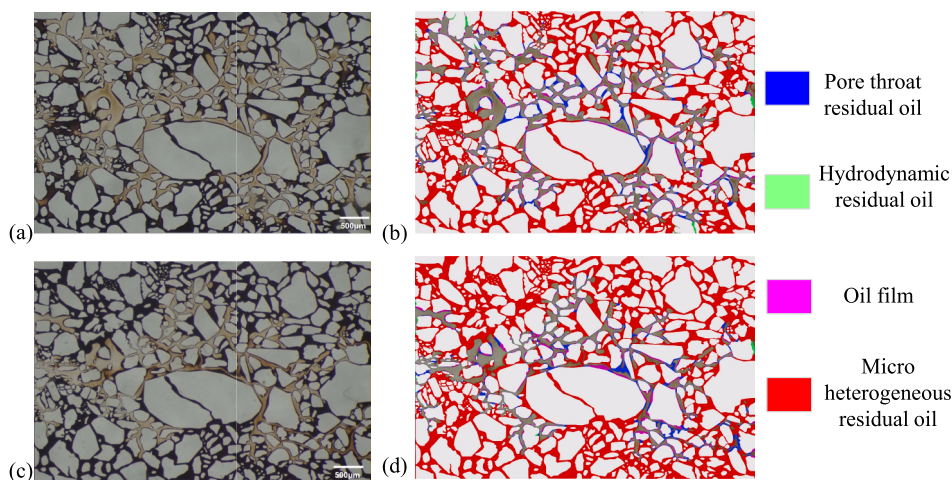


Fig. 7. Microscopic oil displacement results with CDs and SDBS flooding. Residual oil images after (a) G-CDs with SDBS flooding and (c) B-CDs with SDBS flooding. The distribution of various types of classified residual oils after (b) G-CDs with SDBS flooding and (d) B-CDs with SDBS flooding.

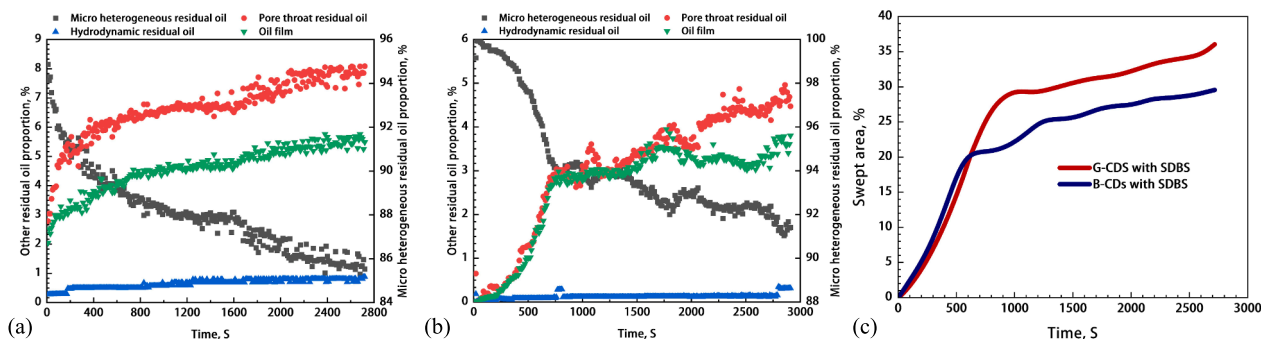


Fig. 8. The relative amount changing of four type residual oils in (a) G-CDs with SDBS flooding and (b) B-CDs with SDBS flooding. (c) Comparison of swept areas in the CDs with SDBS flooding.

obviously affect the wettability alteration ability of G-CDs and B-CDs. The wettability changing may be a more important factor than IFT reduction for nanofluids EOR. The comparison of two CDs on the swept areas in Fig. 8c shows the G-CDs has a swept area of 36.1 %, which is 6.6 % higher than that of B-CDs of 29.6 %. These results have shown that nanoparticles adsorbed at the oil–water interface present better performance for oil mobilization than CDs dispersed in the bulk phase, especially for the mobilization of micro heterogeneous residual oil and oil film.

3.6. Oil displacement performance of two CDs in the porous medium

The core flooding experiments can reflect the oil mobilization in the confined space. The displacing phase in the one-dimensional cylindrical core can sweep the whole areas. The oil recovery equals to the oil displacement efficiency. The oil displacement efficiencies of CDs in Fig. 9a increase with the CDs injection and gradually flatten along with the displacing fluid breakthrough. The oil recovery of G-CDs is lower than that of B-CDs before 1.5 PV but exceeds it afterward. The ultimate oil recovery of G-CDs is 37.5 %, which is 1.7 % higher than that of B-CDs of 35.8 %. Their ultimate oil recoveries are higher than that of SDBS of 32.7 %. The oil recovery difference between SDBS and nanofluid can show the contribution of nanoparticle to oil mobilization. It was around 3 %–5%. Compared to B-CDs, the emulsification can be clearly observed in the effluent during the G-CDs flooding. The differing performance of two CDs in the emulsification in the effluent in Fig. 9b and 9c can explain the oil displacement difference in Fig. 9a. Oil is easily emulsified by the G-CDs due to its interface self-assembly property. The formed emulsions could increase injection pressure through retention and Jamin effect. The higher injection pressure benefits for the swept volume enlargement, contributing to the oil increment. The stable injection pressures at the late period of displacement process indicate there are no nanoparticle precipitation and filtration in the cores.

During the CDs injecting process, the retained oil droplet could not be fully displaced in the initial flooding stage. Therefore, the oil displacement efficiency of G-CDs with the good emulsifying capacity in Fig. 9a is lower than that of B-CDs at the beginning. With the continuous injection of displacing fluid, more retained oil droplets are displaced, achieving a higher unlimited oil displacement efficiency. The cleaner core sample after G-CDs flood in Fig. 9c also indicates the particles adsorbed at oil–water interface presents a better performance in the oil mobilization than particles dispersed in the bulk phase.

It should be noted that the oil recovery difference between B-CDs and G-CDs in the micromodel is larger than that in the core flooding experiments because of the effect of scale. The oil recovery difference usually decreases with the scale upgrading. Compare to the effect of oil displacement efficiency in the core flooding experiments, the additional micro-swept effects in the micromodel will magnify the difference of oil recovery factor between the two nanofluids. But the results of two scales both show the nanoparticles adsorbed at the oil–water interface featured with a higher oil recovery factor.

4. Conclusion

Two fluorescent nanosized CDs, exhibiting distinct affinities for the oil phase and water phase with the help of SDBS, were successfully synthesized to investigate their impact on oil mobilization in the porous medium. Laser confocal images vividly revealed that G-CDs could adsorb on the oil–water interface whereas B-CDs dispersed in the bulk phase. The different locations of the nanoparticles resulted in several macro-phenomenon differences, including; 1) G-CDs demonstrated a strong capability for the wettability changing from oil wet to water wet, 2) it has a lower interface tension with kerosene, and 3) the emulsions formed displayed higher shear viscosity and better stability. These static performances collectively suggest that the particles adsorbed at oil–water interface is more favorable for the practical application.

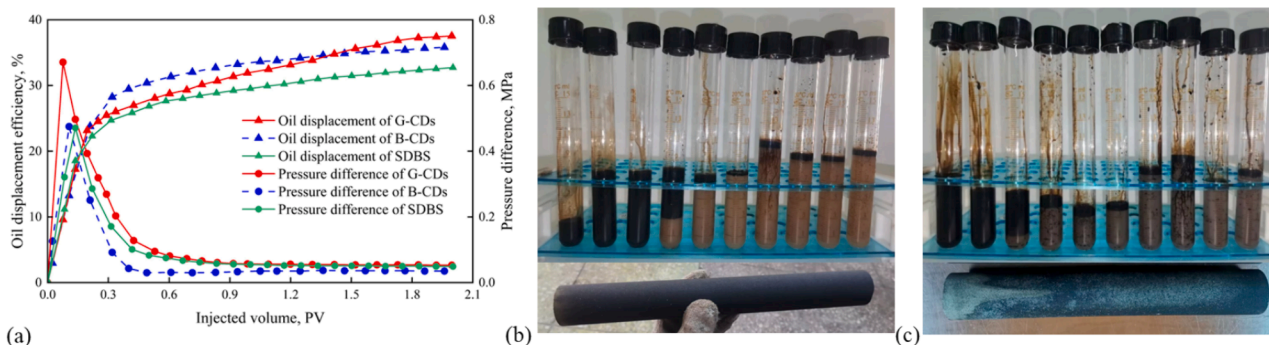


Fig. 9. Oil displacement results of two nanofluids and SDBS (a) oil recovery curve and injection pressure curve of G-CDs with SDBS flooding, B-CDs with SDBS flooding and SDBS flooding (b) effluent and core image after displacement using B-CDs with SDBS and (c) G-CDs with SDBS.

The same conclusion can be drawn from the displacement experiments conducted at both the pore scale and core scale. In comparison to B-CDs, which tend to disperse in the bulk phase, G-CDs adsorbed at the interface demonstrate a superior ability to mobilize oil film and exhibit enhanced emulsification capabilities during the flooding process. These characteristics result in G-CDs covering a larger swept area of 6.6 % in the microfluidic flooding and achieving a higher displacement efficiency of 1.7 % in the core flooding experiments. The oil mobilization performance observed at both scales corroborates that a particle adsorbed at oil–water interface can simultaneously enhance swept volume and improve oil displacement efficiency.

Nanoparticles adsorbed at interface undergo a two-dimensional environment, speculated to impart distinct properties compared to particles dispersed in the bulk phase. Our study validates this conjecture through static performances and dynamic oil displacements using two CDs. The significance of our work lies in clarifying the interfacial property requirement of nanomaterials for practical application in enhanced oil recovery. Notably, particles adsorbed at the oil–water interface are favored for oil mobilization in the oil-wet reservoir, especially for the tight oil recovery. While prior research [54–59] has suggested wettability changes by nanoparticles could improve imbibition efficiency, the mechanism's regarding the need for nanoparticle interface properties remain unreported. Our study emphasizes the importance of considering particles adsorbed at oil–water interface when developing or selecting nanomaterials for EOR. We anticipate that our findings on the properties changing due particles adsorbed interface can extend to application in personal care products, chemical synthesis, and various industries.

CRediT authorship contribution statement

Zheyu Liu: Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Conceptualization. **Jinxin Cao:** Visualization, Software, Investigation. **Li Liu:** Investigation, Data curation. **Yingchun Niu:** Validation, Formal analysis. **Mohammed A Samba:** Writing – original draft, Validation. **Wenxu Wang:** Writing – original draft, Data curation. **Quan Xu:** Visualization, Formal analysis, Conceptualization. **Yiqiang Li:** Supervision, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.molliq.2024.125802>.

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